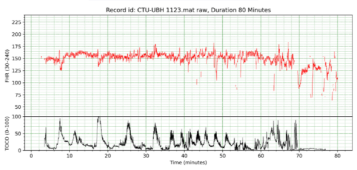
CTGDL - Data Collection

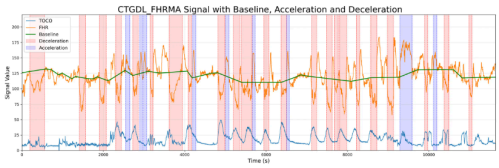
CTGDL: Cardiotocography Merged Dataset

CTU-UHB



CTGDL_CTU_UHB
- **552 cases**
- **682 total hours**
- **74 mean duration**
- **Complete clinical data**

FHRMA



CTGDL_FHRMA
- **137 cases**
- **234 total hours**
- **134 mean duration**
- **Morphological Annotation:** Baseline, Acceleration & Deceleration

SPAM

CTGDL_SPAM
- **297 cases**
- **1527 total hours**
- **309 mean duration**
- **Long duration**

This section provides an overview of the datasets utilized for Cardiotocography (CTG) analysis, focusing on their content, origin, and the format in which they are provided for machine learning pipelines.

SPAM Dataset

The SPaM dataset consists of intrapartum CTG recordings collected for the 2nd Signal Processing and Monitoring (SPaM) in Labour Workshop held in Oxford, UK, in 2The associated CTG Challenge 2017 provided a shared benchmark for methods in fetal heart rate (FHR) and uterine contraction (UC) signal analysis, focusing on labour monitoring, fetal stress detection, and advanced signal processing and classification techniques.

In this repository, the SPaM subset is provided as preprocessed time-series files (FHR and UC at 4 Hz, with missing data preserved via NaN markers where applicable), ready for use in machine learning pipelines for fetal stress prediction and CTG pattern analysis.

Source and documentation:

- Workshop page:
  - https://users.ox.ac.uk/~ndog0178/spam2017.htm
- Data description:
  - https://users.ox.ac.uk/~ndog0178/CTGchallenge2017.htm
  - https://users.ox.ac.uk/~ndog0178/CTGchallenge2017.pdf
- Download (original workshop database):
  - http://ctg.ciirc.cvut.cz/other_files/CTG_workshop_database.zip

FHRMA Dataset

The FHRMA dataset contains fetal heart rate recordings with expert morphological annotations, including baseline, accelerations, and decelerations. It was originally introduced as a resource for training and evaluating algorithms for CTG morphological analysis and enables supervised and transfer-learning approaches for detecting clinically relevant patterns in FHR.

In this repository, FHRMA recordings are standardized to a common sampling rate and exported to tabular formats with aligned annotation channels to facilitate downstream modelling.

CTU-UHB

The CTU-UHB intrapartum CTG database comprises 552 recordings collected at Czech Technical University and University Hospital Brno, with detailed clinical outcome data including umbilical artery pH, Apgar scores, and maternal/neonatal parameters. Each recording includes FHR and UC signals acquired during labour, typically starting up to 90 minutes before delivery, and is accompanied by rich metadata suitable for supervised outcome prediction and retrospective CTG analysis.

In this repository, CTU-UHB data are provided as standardized 4 Hz FHR/UC time series and harmonized metadata tables, enabling consistent integration with SPaM and FHRMA foexperiments.